



Tropical oceans drive decadal trends in surface mass balance over Wilkes Land, East Antarctica

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Abstract

The East Antarctic Ice Sheet, once considered stable, has experienced mass loss in recent decades, with regional disparities in surface mass balance (SMB) trends playing a significant role. This study investigates the multi-decadal SMB trend in Wilkes Land, East Antarctica, utilizing ice core records, modeled reconstructions and reanalysis datasets. We provide robust evidence for a declining trend in SMB over Wilkes Land during 1979–2012, primarily driven by decreasing precipitation. Statistical analysis and atmospheric model experiments reveal that this trend is largely influenced by the multi-decadal warming of the tropical Atlantic and Indian Oceans, while the cooling of the eastern tropical Pacific appears to have a limited impact. The warming of the tropical Atlantic and Indian Oceans generates Rossby waves that propagate towards Wilkes Land, inducing cyclonic anomalies and reducing moisture transport from the ocean to the ice sheet. These findings highlight the crucial role of tropical–polar teleconnections in shaping regional hydroclimate variability in East Antarctica on multi-decadal timescales, reveal a basin-dependent contrast in the tropical drivers of Antarctic climate variability, and have important implications for future projections of Antarctic Ice Sheet mass change.

Keywords East Antarctica · Surface mass balance · Atmospheric teleconnection · Tropical SST

1 Introduction

The Antarctic Ice Sheet is the largest freshwater reservoir on Earth, enough to raise global sea level by 58 m (Morlighem et al. 2020). Of this, the East Antarctic Ice Sheet (EAIS) contains the most ice of the Antarctic Ice Sheet (52 m), while the West Antarctic Ice Sheet (WAIS) contains less (5 m) (Morlighem et al. 2020). Over the past 3 decades, the Antarctic Ice Sheet has been losing mass at an accelerated rate, with most mass loss concentrated in West Antarctica, while the EAIS is considered stable (Schroeder et al. 2019; Stokes et al. 2022), although some marine-based glaciers are losing mass (Li et al. 2023b; Vankova et al. 2023). Recent satellite-based and reconciled mass-balance assessments further suggest that parts of the EAIS have also been losing mass during the satellite era, although the magnitude and spatial extent remain regionally heterogeneous (Otosaka et al. 2023; Rignot et al. 2019). Accelerated glacier flow along the Pacific coast of East Antarctica, driven in part by the intrusion of warm circumpolar deep water, has contributed substantially to mass loss in vulnerable marine-based sectors of the EAIS (Greenbaum et al. 2015; Li et al. 2023a, b; Vankova et al. 2023).

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On the other hand, surface mass balance (SMB) also plays an important role in modulating the mass changes of the EAIS (Akers et al. 2022; Boening et al. 2012; Medley et al. 2018). It is reported that the integrated SMB of the EAIS is expected to increase due to the higher moisture-holding capacity of the atmosphere induced by global warming, keeping the ice sheet in balance over the twenty-first century (Frieler et al. 2015; Palerme et al. 2017; Tewari et al. 2022). However, substantial regional disparities in multi-decadal SMB trends exist across East Antarctica. While SMB in Dronning Maud Land has increased since the twenty-first century (Lenaerts et al. 2013; Medley et al. 2018), Wilkes Land has shown a sustained decrease over the past 4 decades, as indicated by modeled reconstructions (van Wessem et al. 2018; Agosta et al. 2019), contributing to the mass loss of grounded ice and nearby ice shelves there (Li et al. 2023a). This regional SMB decline therefore stands in contrast to the broad thermodynamic expectation of increased Antarctic accumulation under warming (Frieler et al. 2015; Palerme et al. 2017; Tewari et al. 2022). By reducing the accumulation component of the EAIS mass budget, such a decline may weaken the compensating effect of SMB against dynamic ice loss (Rignot et al. 2019; Otsaka et al. 2023; Kim et al. 2024).

Recent studies reveal that tropical oceans are key drivers of regional climate variations in Antarctica. On interannual timescales, the El Niño–Southern Oscillation (ENSO) (Turner 2004) is closely linked to surface temperature variability in the Antarctic Peninsula and West Antarctica (Ding and Steig 2013) and sea-ice concentration in the Amundsen and Bellingshausen seas (Nuncio and Yuan 2015; Yuan and Martinson 2001). Recent studies have further highlighted ENSO-related climate variability as an important contributor to Antarctic ice-mass change and ice-shelf variability (King et al. 2023; Zinck et al. 2025). By contrast, the Indian Ocean Dipole (IOD) (Saji et al. 1999) has also been linked to Antarctic climate variability through its modulation of atmospheric circulation and moisture transport (Marshall et al. 2017; Donat-Magnin et al. 2020). In recent decades, the tropical Atlantic and Indian Oceans have warmed substantially (Li et al. 2014; Zhang et al. 2018; Lee et al. 2015), while the eastern tropical Pacific has cooled (Kosaka and Xie 2013; Li et al. 2016). The sea surface temperature variations related to the phase change of the Interdecadal Pacific Oscillation (IPO) (Henley et al. 2015; Zhang et al. 1997) and the Atlantic Multidecadal Oscillation (AMO) (Schlesinger and Ramankutty 1994) impact surface temperature and sea ice in the West Antarctic (Donat-Magnin et al. 2020; Li et al. 2021; Meehl et al. 2019).

Previous work has established a dynamical framework in which Antarctic snowfall and SMB variability are shaped not only by thermodynamic moisture availability, but also by

large-scale circulation anomalies and the associated moisture-transport pathways (Marshall et al. 2017; Thomas et al. 2015; Bodart and Bingham 2019; Donat-Magnin et al. 2020; Lee et al. 2025; Macha et al. 2025). Southern Hemisphere circulation modes, including the Southern Annular Mode (SAM) and Pacific–South American (PSA) patterns, exert regionally distinct influences on Antarctic precipitation, indicating that the response of snowfall to remote forcing is strongly sector-dependent (Marshall et al. 2017). Tropical SST variability can project onto these circulation modes through Rossby-wave teleconnections. ENSO-related forcing, for example, has been linked to the Amundsen Sea Low (ASL) variability, moisture transport, and snowfall or SMB anomalies over West Antarctica and the Antarctic Peninsula (Bodart and Bingham 2019; Donat-Magnin et al. 2020; Lee et al. 2025; Macha et al. 2025; Thomas et al. 2015). Man et al. (2025) further showed that twentieth-century tropical Pacific and Atlantic SST variability contributed to dipole-like snowfall trends across West Antarctica. Together, these studies suggest that tropical forcing can influence Antarctic SMB, but its regional impact depends on the basin of SST forcing and the associated atmospheric teleconnection pathway. For Wilkes Land, it remains unclear whether the observed SMB decline is linked mainly to the traditionally emphasized Pacific influence, or whether Atlantic and Indian Ocean variability also play important roles. This is especially important because the role of Indian Ocean variability in East Antarctic precipitation has recently begun to receive attention and remains less well quantified than the Pacific and Atlantic influences (Ma et al. 2025). Here, we address this question by combining ice-core constraints, gridded SMB and precipitation products, and basin-specific atmospheric model experiments to examine whether the recent Wilkes Land SMB decline can be linked to tropical SST forcing, how the influence differs among the Pacific, Atlantic, and Indian Oceans, and through which Rossby-wave and moisture-transport pathway this influence occurs.

The rest of this study is structured as follows. Section 2 describes the datasets, statistical methods, teleconnection diagnostics, and atmospheric modeling experiments used in this study. Section 3 elucidates the multi-decadal SMB trends observed in Wilkes Land, highlighting the influential role of tropical oceans in driving these trends and the underlying mechanisms at play. Section 4 discusses these findings in the context of previous studies, highlights the contrasting roles of different tropical ocean basins, and addresses the broader implications and remaining uncertainties. Section 5 concludes with a summary of the main findings.

2 Data and methods

2.1 Data

2.1.1 Ice-core SMB records

We use the Law Dome ice-core SMB reconstruction from Roberts et al. (2015; Dome Summit South, 66.77° S, 112.81° E) as an annually resolved observational benchmark to complement the spatially continuous model and reanalysis products used below. Derived from four ice cores drilled at the Dome Summit South site, including one long core and three shorter auxiliary cores, this reconstruction extends to 2012 because the uppermost core was drilled in that year (Roberts et al. 2015). We therefore analyze the 1979–2012 segment to match the satellite-era gridded products used elsewhere in this study. Continuous, well-dated ice-core records spanning this full analysis period remain sparse in Wilkes Land and surrounding areas. In particular, several shallow ice cores from nearby regions, such as D66 and GV2, do not extend sufficiently far into the recent decades to robustly resolve a multi-decadal trend over the full 1979–2012 period (Favier et al. 2013; Wang et al. 2021). We therefore also consider the shallow ice-core record from Aurora Basin North (ABN; 71° S, 111° E) as an additional point of comparison (Servettaz et al. 2020).

2.1.2 Regional climate model and reanalysis products

To provide spatially continuous information on SMB and its components, we use outputs from two regional climate models, RACMO2.3p2 (van Wessem et al. 2018) and MARv3.6.4 (Agosta et al. 2019), together with the fifth-generation European Centre for Medium-Range Weather Forecasts reanalysis (ERA5) (Hersbach et al. 2020). Given the severe scarcity of continuous in situ observations across Antarctica, relying on a single gridded product can introduce substantial structural uncertainties (Mottram et al. 2021). Specifically, global reanalyses like ERA5 are limited by their relatively coarser resolution over complex coastal topography and their lack of sophisticated snowpack processes, such as blowing snow sublimation (Gossart et al. 2019). Conversely, while regional climate models (RACMO2 and MAR) are explicitly tailored for polar regions, their distinct parameterizations of surface processes and microphysics can still yield varying regional SMB gradients (Agosta et al. 2019; van Wessem et al. 2018). Therefore, we cross-reference these diverse datasets to ensure robust assessments. From RACMO2 and MAR, we analyze SMB together with its major components, including precipitation, sublimation, and runoff, over 1979–2012. For ERA5, because Antarctic SMB is dominated by snowfall and sublimation over most

grounded-ice regions, we use net precipitation (precipitation minus sublimation) as a proxy for SMB variability, following recent Antarctic climatological assessments (Medley and Thomas 2019; Wang et al. 2020b).

ERA5 is also used to diagnose atmospheric circulation, moisture transport, and precipitation-related changes over Wilkes Land. Specifically, we analyze geopotential height at 200 hPa, sea-level pressure, horizontal wind velocity, specific humidity, and surface pressure to diagnose large-scale circulation anomalies, Rossby-wave-related teleconnections, and vertically integrated moisture transport associated with Wilkes Land SMB variability and tropical SST forcing. The gridded products are analyzed on their native analysis grids for the trend maps and regression analysis.

2.1.3 Topography and SST datasets

Antarctic surface elevation is taken from BedMachine Antarctica, version 3 (Morlighem et al. 2020). The original 500 m × 500 m topography is regridded to the spatial resolution of ERA5 and CAM5, respectively, and is used to calculate the local gradient direction of surface elevation for diagnosing the onshore component of moisture transport relevant to orographic snowfall.

Monthly SST fields are taken from the Hadley Centre Sea Ice and Sea Surface Temperature dataset (HadISST) at 1° × 1° resolution (Rayner et al. 2003). HadISST is used for three purposes: (1) estimating multi-decadal SST trends during 1979–2012, (2) constructing basin-scale tropical SST indices, and (3) prescribing lower-boundary SST forcing in the atmospheric model experiments.

2.2 Trend estimation and regression analyses

Given the high interannual variability and potential non-normal distribution characteristic of Antarctic snow accumulation and climate data (Thomas et al. 2017), we estimate linear trends using the non-parametric Sen's slope estimator (Gocic and Trajkovic 2013), which is robust to outliers and non-normality. Statistical significance of the trends is assessed using the Mann–Kendall test (Gocic and Trajkovic 2013). Uncertainty ranges for trend estimates are obtained by bootstrap resampling (Tibshirani and Efron 1994).

To diagnose circulation, precipitation, and moisture-transport anomalies associated with Wilkes Land SMB variability and tropical SST indices (defined below), we utilize ordinary least-squares linear regression on detrended and annual-mean time series. Sensitivity tests using non-detrended annual series yield the same sign of these relationships and similar teleconnection patterns. Crucially, because atmospheric and oceanic time series often exhibit strong autocorrelation that can artificially inflate statistical significance in polar

teleconnection studies (Li et al. 2014), the statistical significance of all regression coefficients is evaluated using a two-tailed Student's *t*-test with effective degrees of freedom strictly adjusted for temporal autocorrelation (Bretherton et al. 1999).

For the ice-core-based regression analyses, the Law Dome SMB series are normalized by their standard deviation prior to regression so that the resulting coefficients can be interpreted as anomalies per one standard deviation of SMB variability. For the basin-index regressions, SST indices are detrended unless otherwise noted. Because the eastern tropical Pacific exhibited cooling over 1979–2012, we use the negative Niño-3.4 index so that positive values consistently correspond to the observed sign of the multi-decadal Pacific anomaly during the analysis period.

To quantify tropical teleconnections and their associated atmospheric pathways, we regress SMB-related fields and key atmospheric variables over Antarctica onto the tropical SST indices defined below (negative Niño-3.4, TA20, and IOBM). This approach characterizes the precipitation and circulation anomalies over Wilkes Land and the wider Southern Hemisphere associated with SST variability in each basin. The regressed fields include SMB, precipitation, geopotential height at 200 hPa, sea-level pressure, and vertically integrated moisture transport, depending on the diagnostic purpose. The statistical significance of these regression coefficients is assessed using the same procedure described above.

2.3 Teleconnection diagnostics

2.3.1 Rossby-wave activity flux

Following established methodologies for diagnosing tropical-to-Antarctic atmospheric teleconnections (Ding et al. 2011; Li et al. 2021), we calculate the Takaya–Nakamura (T–N) wave activity flux (Takaya and Nakamura 2001). The T–N wave activity flux diagnoses the propagation of quasi-stationary Rossby waves, and its horizontal components are given by

$$F = \frac{p}{2|\bar{\mathbf{U}}|} \left\{ \begin{array}{l} \bar{u}(\psi'^2_x - \psi'\psi'_{xx}) + \bar{v}(\psi'_x\psi'_y - \psi'\psi'_{xy}) \\ \bar{u}(\psi'_x\psi'_y - \psi'\psi'_{xy}) + \bar{v}(\psi'^2_y - \psi'\psi'_{yy}) \end{array} \right\} \quad (1)$$

where overbar denotes the climatological value and prime denotes interannual anomalies. *u* and *v* denote the zonal and meridional wind, respectively. $|\bar{\mathbf{U}}|$ denotes the magnitude of the climatological horizontal wind. $\psi = \frac{\phi}{f}$ is the stream function, where ϕ is the geopotential height and *f* is the Coriolis parameter. *p* is the normalized pressure (pressure/1000 hPa). The subscripts *x* and *y* represent the zonal and meridional gradients, respectively. The horizontal components of the flux are calculated from anomalous streamfunction fields derived

from 200 hPa geopotential height. The resulting vectors indicate the direction and relative magnitude of stationary wave propagation associated with basin-scale SST variability. We use the T–N formulation rather than the Plumb flux (Plumb 1985; Karoly 1989) because it is more appropriate for diagnosing the propagation of quasi-stationary Rossby-wave packets on a spatially varying basic flow (Takaya and Nakamura 2001), which is important for Southern Hemisphere teleconnections extending from the tropics into high latitudes (Li et al. 2015a).

2.3.2 Vertically integrated moisture transport

Vertically integrated moisture transport is calculated from monthly mean wind and specific humidity as

$$Q = \frac{1}{g} \int_{p_t}^{p_s} q \mathbf{V} dp \quad (2)$$

where *q* is specific humidity, *V* is horizontal wind velocity, *p_s* is surface pressure, *p_t* is the upper integration limit, and *g* is gravitational acceleration. In this study, *p_t* is set to 10 hPa, consistent with the atmospheric-column integration available in ERA5 and CAM5.

To quantify the part of the moisture transport most relevant to snowfall over the ice sheet, we project the vertically integrated moisture-transport vector onto the local gradient direction of surface elevation. Positive projected values indicate moisture transport directed upslope toward regions of increasing surface elevation inland, which is favorable for orographic ascent and snowfall, whereas negative projected values indicate downslope or offshore moisture transport that is less favorable for snowfall accumulation (Bromwich 1988). This topography-following projection method has been shown to be critical for accurately attributing coastal Antarctic precipitation anomalies to synoptic and large-scale forcing (Wille et al. 2019). The surface-elevation gradient is calculated from BedMachine surface elevation after bilinear interpolation to the ERA5 or CAM5 grid. So the gradient represents the resolved topographic slope at the atmospheric-model/reanalysis grid scale rather than unresolved local coastal slopes.

2.3.3 Tropical SST indices

We define three tropical SST indices to represent basin-scale variability in the Pacific, Atlantic, and Indian Oceans. The Niño-3.4 index is calculated as the area-mean SST over 5° S–5° N, 170°–120° W and is used to represent tropical Pacific variability (Trenberth and Stepaniak 2001). The tropical Atlantic index (TA20) is defined as the area-mean SST over 20° S–20° N in the Atlantic basin (Li et al. 2014). Tropical

Indian Ocean variability is represented by the Indian Ocean Basin Mode (IOBM) index, defined as the area-mean SST over 20° S–20° N in the Indian Ocean basin (Xie et al. 2009).

2.4 Modeling experiments

2.4.1 Model configuration

To test the causal influence of tropical SST forcing on precipitation over Wilkes Land, we perform atmospheric simulations using the Community Atmosphere Model version 5 (CAM5), developed by the National Center for Atmospheric Research (NCAR). This experimental design has been widely utilized and validated in recent Antarctic teleconnection studies (Ding et al. 2011; Li et al. 2014). CAM5 is run on a finite-volume grid with a horizontal resolution of $1.9^\circ \times 2.5^\circ$ and 26 vertical levels extending from the surface to 3.5 hPa. Over land, surface heat and moisture fluxes are provided by the Community Land Model (CLM). Sea surface temperature and sea-ice coverage are prescribed boundary conditions in these atmospheric simulations. External forcings such as atmospheric CO₂ and aerosols are fixed at climatological mean values. So, these simulations isolate the atmospheric response to prescribed SST variability and do not include direct radiative-forcing effects or aerosol-driven circulation changes.

2.4.2 Construction of SST forcing anomalies

We conduct three groups of basin-forcing experiments to isolate the effects of SST variability in the Pacific, Atlantic, and Indian Oceans. The forcing domains are defined as 40° S–65° N for the Pacific, 30° S–75° N for the Atlantic, and 25° S–25° N for the Indian Ocean. Outside the target basin, SSTs are prescribed as the 1979–2012 monthly climatology with a repeating annual cycle.

Within each target basin, prescribed SST perturbations are constructed from HadISST as the observed monthly anomalies during 1979–2012 relative to the corresponding 1979–2012 monthly climatology. These anomalies are imposed directly, without linear detrending, standardization, or amplitude scaling prior to application. To avoid spurious discontinuities at the basin boundaries, the imposed SST anomalies are smoothly tapered to zero within a 10° latitude–longitude buffer zone surrounding each forcing region, thereby ensuring a continuous transition to the climatological background state outside the forcing domain. Sea-ice coverage is prescribed from the same boundary dataset used to force the atmospheric simulations. The simulated atmospheric response therefore reflects the imposed basin-scale SST anomalies under prescribed oceanic and sea-ice boundary conditions.

2.4.3 Experiment length, spin-up, and ensembles

Each basin-forcing experiment consists of 12 ensemble members. Ensemble averaging is used to suppress atmospheric internal variability and isolate the SST-forced response. Prior to the perturbation experiments, a control simulation was integrated under climatological boundary conditions, and the basin-forcing experiments were then branched from this control run. Specifically, the 12 ensemble members were initialized from different years and dates sampled from the control simulation and integrated continuously from 1979 to 2012 under the prescribed basin-scale SST perturbations. The results presented in this study are based on the ensemble-mean response over the full 1979–2012 integration period.

2.4.4 Diagnosis of basin-specific precipitation responses

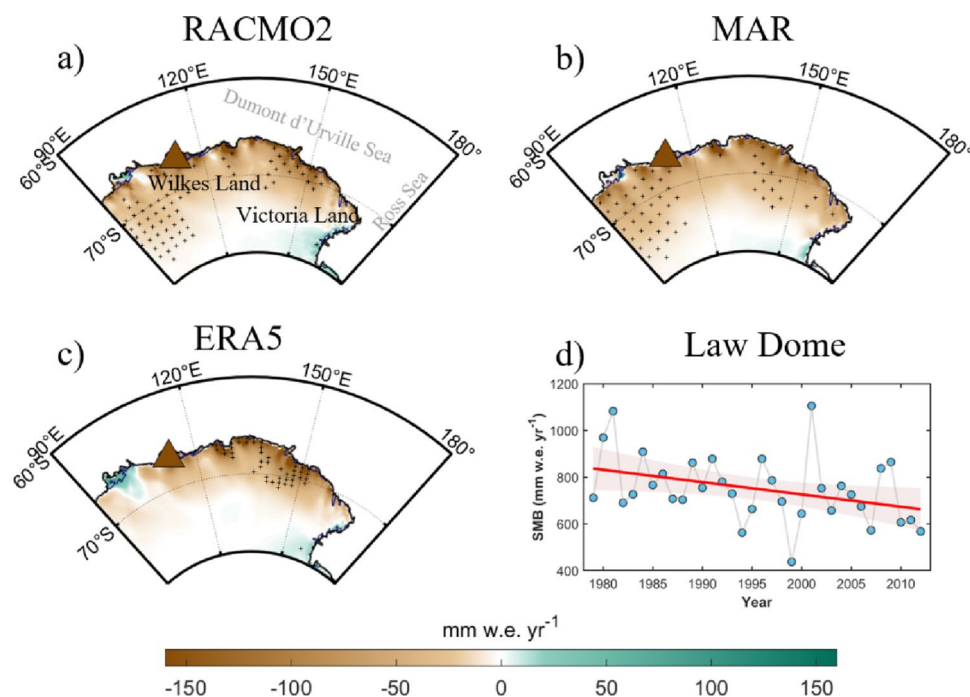
To compare model-simulated and observed relationships, we regress simulated precipitation from each basin experiment onto the corresponding tropical SST index (negative Niño-3.4, TA20, or IOBM). These regression patterns are then compared with analogous regressions derived from ERA5 and the ice-core SMB record. Agreement between the basin-forced CAM5 experiments and the observationally based regressions is interpreted as evidence that SST variability in that basin makes a causal contribution to the precipitation anomalies over Wilkes Land.

3 Results

3.1 SMB decline over Wilkes Land and its regional representativeness

To assess multi-decadal SMB changes over Wilkes Land during 1979–2012, we synthesize evidence from the Law Dome ice-core record, regional climate model reconstructions, and reanalysis-based diagnostics (Fig. 1). We use the Law Dome SMB reconstruction as a long, well-dated coastal benchmark for variability in the Wilkes Land sector. The Law Dome record exhibits a significant decreasing trend of -173.9 ± 157.2 mm w.e. yr⁻¹ during this period (Fig. 1d). While a single site may not fully capture spatial heterogeneity across the catchment (Cavitt et al. 2023; Richardson and Holmlund 1999), its broader regional relevance is supported by the Aurora Basin North shallow ice-core record, which shows a slight concurrent decrease in snow accumulation (Fig. S1; -13.8 mm w.e. yr⁻¹), although this trend is weak and subject to considerable uncertainty. Although the wider compilation of SMB observations across Wilkes Land remains sparse and most available shallow ice cores are too

Fig. 1 SMB trends over Wilkes Land and Victoria Land in East Antarctica. **a** shows the SMB trends during 1979–2012 from the RACMO2 dataset. Black crosses denote the areas where the trends are statistically significant at the 90% confidence level. The triangle in **(a)** indicates the location of the Law Dome ice core, with the color indicating the linear SMB trend in the Law Dome ice core. **b** and **c** are the same as **(a)**, but show the SMB trends from the MARv3 dataset **(b)** and ERA5 **(c)**. **d** shows the SMB time series from the Law Dome ice core. Light blue circles connected by the gray line represent the annual snow accumulation variations. The solid red line denotes the linear regression trend over the study period. The light red shaded area indicates the 95% confidence interval of the linear fit



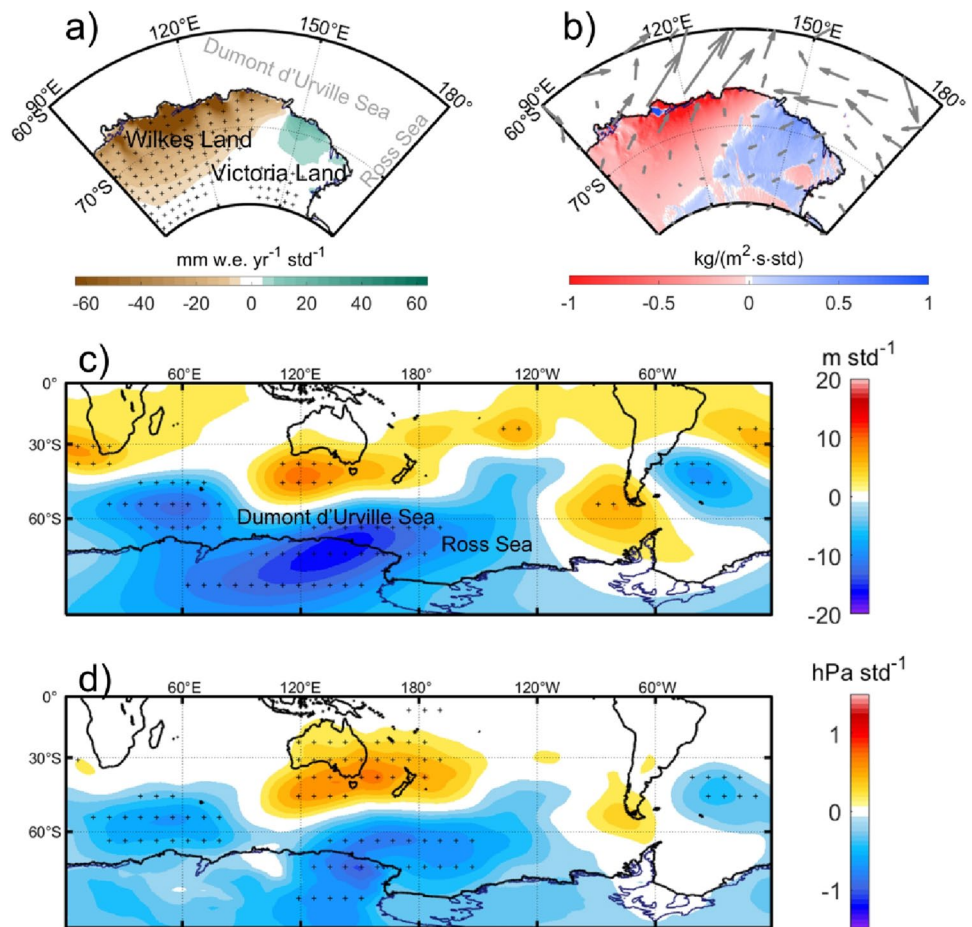
short to robustly resolve multi-decadal trends (Favier et al. 2013; Wang et al. 2021), spatial consistency is corroborated by gridded products. Consistent with this decline, RACMO2 and MAR show negative SMB trends across Wilkes Land and Victoria Land (Fig. 1a, b), exceeding $20 \text{ mm w.e. yr}^{-1}$ in most areas in RACMO2 and $30 \text{ mm w.e. yr}^{-1}$ in MAR, with coastal trends larger than $150 \text{ mm w.e. yr}^{-1}$ in both datasets. ERA5-derived net precipitation also shows negative trends broadly consistent with the regional climate model results (Fig. 1c).

While RACMO2, MAR, and ERA5 consistently capture the broad multi-decadal SMB decline over Wilkes Land, notable discrepancies exist in the trend magnitudes and fine-scale spatial patterns (Fig. 1). Spatially, the regional climate models (RACMO2 and MAR) exhibit widespread and statistically significant negative trends across the Wilkes Land interior (Fig. 1a, b). In contrast, the negative trend in ERA5 is generally weaker in magnitude and less spatially coherent, with isolated patches of positive trends near the steep coastal margins (Fig. 1c). These discrepancies largely stem from the different spatial resolutions and physical parameterizations of the datasets. As a global reanalysis, ERA5 has a relatively coarser effective resolution, which limits its ability to accurately resolve the steep orographic lifting and complex coastal topography of East Antarctica (Gosart et al. 2019). Furthermore, ERA5 does not fully incorporate complex snowpack processes, such as blowing snow sublimation—a critical component of the Antarctic coastal mass budget (Agosta et al. 2019). Conversely, RACMO2 and MAR are specifically tailored for polar regions with sophisticated multilayer snow models. RACMO2 is known

to explicitly resolve drifting snow erosion and sublimation, which often leads to more pronounced and realistic coastal mass loss (van Wessem et al. 2018). MAR also employs advanced cloud microphysics and boundary layer schemes but can sometimes yield slightly different precipitation gradients due to its specific moisture advection handling (Mottram et al. 2021). Given the cold conditions over Wilkes Land, interannual-to-decadal SMB variability is expected to be primarily controlled by snowfall (Fig. S2); accordingly, we use net precipitation as a first-order proxy for SMB here, and precipitation as the main diagnostic variable in the following analyses. We note that this approximation is less accurate in steep coastal regions where blowing-snow processes, sublimation, melt, and unresolved orographic gradients can locally affect SMB, although it remains a reasonable first-order approximation at larger spatial scales. Taken together, these datasets support the presence of a robust regional SMB decline over Wilkes Land during 1979–2012, while also indicating spatial variability in trend magnitude across the broader sector.

The regional representativeness of the Law Dome record is further supported by regression of ERA5 net precipitation onto the Law Dome SMB series (Fig. 2a). Positive regression coefficients are concentrated over Wilkes Land, indicating that years of reduced SMB at Law Dome are associated with reduced precipitation over the broader Wilkes Land sector. By contrast, the corresponding signal over Victoria Land is weak and statistically insignificant. These results suggest that the Law Dome record captures variability relevant to Wilkes Land more strongly than to adjacent

Fig. 2 Regression patterns of precipitation, moisture transports and atmospheric circulation on the SMB in the Law Dome ice core. **a** shows the pattern of regression coefficients of precipitation in ERA5 on the normalized and detrended SMB in the Law Dome ice core. Black crosses denote the areas where the regression coefficients are statistically significant at the 95% confidence level. **b** shows the regression pattern of vertically integrated moisture transports (vectors) and their projection onto the gradient direction of surface elevation (color shading). **c** shows the regression pattern of the 200 hPa geopotential height. **d** shows the regression pattern of sea level pressure



sectors, supporting its use as a coastal proxy for regional SMB change.

Despite these known systematic biases and differences in magnitude, the consensus among these independent datasets on the overarching negative sign of the SMB trend provides strong confidence in the regional drying signal. The agreement between the high-resolution RCMs, the global reanalysis, and the Law Dome ice-core record robustly confirms that the Wilkes Land SMB decline is a physical reality rather than an artifact of a single dataset. Having established a robust SMB decline over Wilkes Land and demonstrated sectoral coherence with the Law Dome constraint, we next diagnose the large-scale atmospheric circulation and moisture-transport signatures associated with the reduced SMB in this region.

3.2 Circulation and moisture-transport signature linked to low SMB

To retrieve the atmospheric circulation pattern associated with the decreasing SMB in Wilkes Land, we regress the geopotential height at the 200 hPa level and sea level pressure on the negative normalized SMB from the Law Dome

ice core records. Geopotential height at the 200 hPa level (Fig. 2c) and sea level pressure (Fig. 2d) both show significant low-pressure anomalies across Victoria Land to Dumont d'Urville Sea, manifested as a barotropic cyclonic anomaly. Specifically, one standard deviation decrease in SMB corresponds to about 1 hPa of anomalous surface low-pressure center over Dumont d'Urville Sea. This atmospheric circulation adjustment may influence the regional moisture transports and thus induce the anomalous precipitation over Wilkes Land.

Given that Antarctic snowfall is tightly constrained by the moisture-flux convergence imposed by circulation anomalies, we diagnose the corresponding vertically integrated moisture transport response (Fig. 2b). We regress the vertically integrated moisture transport onto the negative normalized SMB in the Law Dome ice core records (vectors in Fig. 2b), as well as its projection on the gradient direction of the surface elevation of the ice sheet (color shading in Fig. 2b), representing its potential capacity to induce snowfall due to the orographic lifting and adiabatic cooling (Bromwich 1988). Results show that the low-pressure anomaly over Victoria Land and Dumont d'Urville Sea mainly induces the dry and cold wind to flow from inland

towards ocean over Wilkes Land. The regression pattern of moisture transports resembles the precipitation regression pattern in Wilkes Land (Fig. 2a), suggesting that it is the barotropic low-pressure anomaly over Victoria Land and Dumont d'Urville Sea that induces the anomalous moisture transports from Wilkes Land to Dumont d'Urville Sea, resulting in the decreasing trend of the precipitation and subsequent SMB in Wilkes Land.

The low-pressure anomaly over Victoria Land and Dumont d'Urville Sea, corresponding to the decreasing SMB in Wilkes Land, is also accompanied by high-pressure anomaly over the Drake Passage and low-pressure anomalies over subtropical Atlantic and Indian sectors of Southern Ocean. The anomalous circulation pattern manifests as a Rossby wave train traversing the Southern Hemisphere, implying the influence of remote forcing from tropical oceans through atmospheric teleconnections. Below, we further investigate the contributions of tropical oceans in shaping the SMB trends in Wilkes Land.

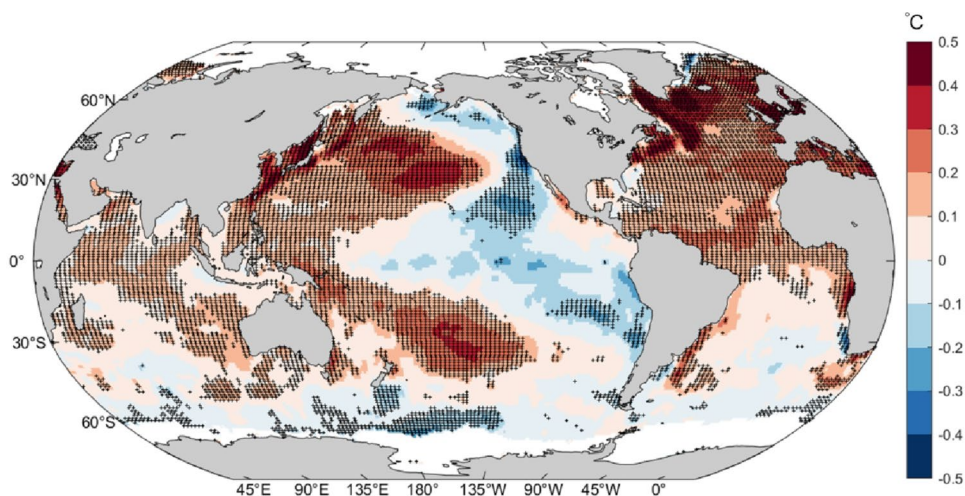
3.3 Linkages between precipitation changes and tropical SST

During 1979–2012, the tropical oceans experienced pronounced and contrasting multi-decadal SST changes across the three tropical ocean basins (Fig. 3). In the tropical Pacific, the IPO shifts from a positive phase to a negative phase, resulting in a moderate cooling of 0.38 ± 0.87 °C (in the Niño-3.4 region). Meanwhile, the Atlantic experiences a shift of AMO from a negative phase to a positive phase, leading to a significant warming trend of 0.44 ± 0.41 °C in the tropical Atlantic (20° S– 20° N) (Fig. 3). For the Indian Ocean, the tropical SST (20° S– 20° N) also shows significant increase of 0.31 ± 0.23 °C. These basin-scale SST changes provide a plausible source of remote forcing that can modulate Southern Hemisphere circulation and, in turn, precipitation and SMB over coastal East Antarctica

via atmospheric teleconnections. While the tropical Pacific's influence on Antarctic climate—particularly over West Antarctica and the Amundsen Sea—has been extensively documented in the literature (Ding et al. 2011; Turner et al. 2016), the relative roles of the three tropical ocean basins in driving hydroclimate changes over East Antarctica remain less understood. Recent studies have increasingly highlighted the Atlantic and Indian Oceans as potent drivers of high-latitude atmospheric teleconnections (Li et al. 2014; Purich and England 2019), prompting us to systematically evaluate their respective contributions to Wilkes Land SMB decline.

To evaluate the basin-specific association between tropical SST variability and hydroclimate over Wilkes Land, we regress ERA5 precipitation and the Law Dome ice-core SMB record onto the tropical SST indices in the three basins (negative Niño-3.4, TA20, and IOBM; Fig. 4a, c and e). The regression results reveal a clear contrast among basins: the Pacific cooling contributes to a weak decreasing trend in precipitation over Wilkes Land (Fig. 4a), of a much smaller magnitude than the regression results from the Atlantic and Indian Oceans (Fig. 4c, e). Importantly, the ice core regression result shows a small contribution of the tropical Pacific cooling to a decrease in precipitation over Wilkes Land, with a change of < 5 mm w.e. yr^{-1} corresponding to 1 °C of eastern tropical cooling, suggesting a limited influence of the Pacific Ocean on the decreasing SMB over Wilkes Land. In contrast, the tropical Atlantic warming largely contributes to precipitation changes over Wilkes Land, with a decrease of about 20 mm w.e. yr^{-1} corresponding to 1 °C of tropical Atlantic warming revealed by the ice core records (Fig. 4c). The tropical Indian Ocean also contributes to a decrease in SMB over Wilkes Land, with a decrease of about 130 mm w.e. yr^{-1} corresponding to 1 °C of tropical Indian Ocean's warming (Fig. 4e). However, both the Atlantic and Indian Ocean regression patterns show insignificant changes in the SMB over Victoria Land. In summary, the regression results

Fig. 3 Multi-decadal trends in sea surface temperature from 1979 to 2012. Black crosses denote the areas where the trends are statistically significant at 95% confidence level



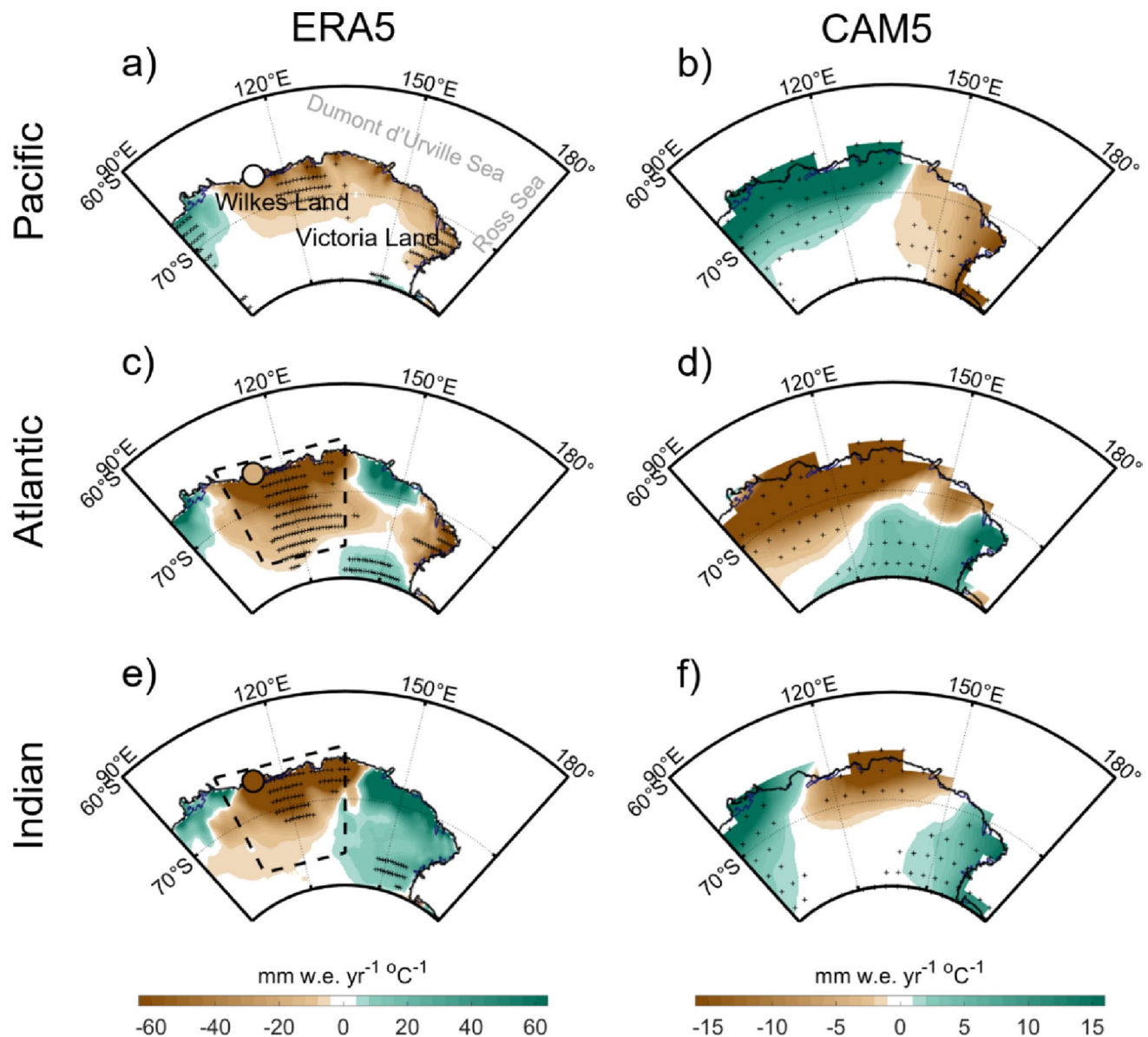


Fig. 4 Regression patterns of precipitation on the tropical SST indices. **a** shows the pattern of regression coefficients of the detrended precipitation on the tropical Pacific index, detrended negative Niño-3.4. The circles mark the regression coefficients of Law Dome SMB onto the corresponding tropical SST indices. Black crosses denote the areas where the regression coefficients are statistically significant at the 95% confidence level. **b** shows the regression pattern of simulated precipitation by CAM5 on negative Niño-3.4. **c** and **d** show the regres-

sion pattern of precipitation in ERA5 (**c**) and simulated precipitation from CAM5 (**d**) on the tropical Atlantic index, detrended TA20. **e** and **f** show the regression pattern of precipitation in ERA5 (**e**) and simulated precipitation from CAM5 (**f**) on the tropical Indian Ocean index, detrended IOBM. The black boxes in (**c**) and (**e**) denote the Wilkes Land target region used to calculate the regional precipitation index for the quantitative correlation analysis

indicate that both the Atlantic and Indian Oceans contribute to the decreasing trends in SMB over Wilkes Land.

While the observational regressions indicate a strong statistical link between the Atlantic/Indian Oceans and Wilkes Land SMB, statistical methods alone cannot definitively isolate causality due to the strong covariance among tropical ocean basins. To cleanly separate the remote impacts of each basin, we conduct three atmospheric model experiments (CAM5), each forced by SST anomalies from a

single ocean basin (Pacific, Atlantic, and Indian Oceans). By regressing the simulated precipitation from these experiments onto their respective SST indices (negative Niño-3.4, TA20, and IOBM), we retrieve the precipitation response to tropical SST changes in each ocean basin (Fig. 4b, d and f). Results show that the tropical Pacific cooling induces a decrease in precipitation in Victoria Land but an increase in precipitation in Wilkes Land (Fig. 4b). However, the Atlantic Ocean and Indian Ocean both induce decreasing

precipitation in Wilkes Land but an increasing precipitation in Victoria Land (Fig. 4d, f), which resemble the regression pattern derived from the ice-core record and ERA5.

In summary, the consistency between the regression analysis and the modeling results shows that the decreasing trends in SMB in Wilkes Land are mainly induced by warming in the tropical Atlantic and Indian Oceans, while cooling in the eastern tropical Pacific Ocean plays a limited role. These findings build upon emerging evidence that the Atlantic and Indian basins play a critical, and sometimes dominant, role in East Antarctic climate variability, distinct from the Pacific-dominated signals in West Antarctica. Having identified the tropical Atlantic and Indian Oceans as the primary causal forcing, the next crucial step is to elucidate the specific physical pathways and atmospheric dynamics that bridge these remote tropical oceans to the coastal EAIS.

3.4 Mechanisms: Rossby-wave teleconnections and moisture-transport modulation

Given that quasi-stationary Rossby wave trains provide an efficient pathway for tropical-to-polar teleconnections (Ding and Steig 2013; Simpkins et al. 2014), we next diagnose the large-scale circulation anomalies associated with basin-scale tropical SST variability and link them to moisture transport and precipitation changes over Wilkes Land. Specifically, we regress ERA5 geopotential height at 200 hPa (Z200) and sea-level pressure (SLP) onto the detrended tropical SST indices (negative Niño-3.4, TA20, and IOBM) (Fig. 5), and use the T–N wave activity flux to infer the propagation of the associated stationary wave activity.

Results show that the Pacific Ocean is associated with an anticyclonic anomaly at the 200 hPa level over Dumont d'Urville Sea and a cyclonic anomaly over Victoria Land and Ross Sea (Fig. 5a). The pattern of sea level pressure regression coefficients exhibits a clear high-pressure ridge over Wilkes Land, especially the Law Dome, suggesting the barotropic features of the atmospheric adjustments induced by the eastern Pacific cooling (Fig. 5b). The associated wave activity flux suggests that the circulation anomalies around Wilkes Land are connected to a zonally propagating Rossby wave train along the Southern Ocean storm-track latitude band, consistent with a remotely forced teleconnection rather than a purely local atmospheric fluctuation.

In contrast, both the tropical Atlantic and Indian Ocean indices are associated with circulation anomalies that favor reduced snowfall over Wilkes Land. Tropical Atlantic warming excites a Rossby wave train traversing the Southern Ocean, culminating in an anomalous, barotropic low-pressure center over Victoria Land and the Dumont d'Urville Sea (Fig. 5c, d). Similarly, warming in the tropical Indian Ocean generates a strong barotropic low-pressure

anomaly immediately east of Law Dome (Figs. 5e, f). In both basins, the wave activity flux indicates a coherent wave train across the Southern Ocean that culminates in the Dumont d'Urville Sea low-pressure anomaly, providing a dynamical bridge between tropical SST variability and East Antarctic circulation changes. These circulation patterns are dynamically consistent with the SMB-linked circulation diagnosed from the ice core record. This agreement supports the interpretation that tropical Atlantic and Indian Ocean variability contributes to Wilkes Land SMB decline by preferentially exciting the same large-scale circulation mode that is observed to accompany low-SMB periods.

To quantitatively assess this teleconnection pathway, we isolate the magnitude of the terminal Rossby waves by defining regional circulation indices based on the respective anomalous cyclonic centers identified in the regression maps. Given the slight spatial offset in the atmospheric responses, we define two regional Z200 indices: one targeting the Atlantic-forced low-pressure center (135° E–205° E, 63° S–73° S; Fig. 5c), and another targeting the more northerly Indian-forced low-pressure center (145° E–195° E, 55° S–65° S; Fig. 5e). Correlation analysis reveals a robust quantitative link between the tropical drivers, these local wave magnitudes, and the resultant precipitation. The Z200 anomaly over the Atlantic-forced center is negatively correlated with the tropical Atlantic SST ($r = -0.30$, $p < 0.1$). Similarly, the Z200 anomaly over the Indian-forced center is significantly negatively correlated with the tropical Indian Ocean SST ($r = -0.34$, $p < 0.05$). This confirms that warming in these basins systematically deepens the regional low-pressure systems off the East Antarctic coast. Furthermore, the magnitudes of both low-pressure anomalies exhibit strong positive correlations with Wilkes Land precipitation anomalies (105° E–135° E, 75° S–65° S; $r = 0.56$, $p < 0.05$; and $r = 0.55$, $p < 0.05$, respectively). This chain of quantitative evidence provides further support for the proposed teleconnection pathway linking basin-scale SST variability to Rossby-wave-induced circulation anomalies and the associated precipitation decline over Wilkes Land.

To further assess the dynamical consistency between the observational regressions and the basin-forced responses, we analyze the atmospheric circulation anomalies simulated in the three sensitivity experiments. Specifically, Rossby-wave-related circulation adjustments are diagnosed by regressing 200-hPa geopotential height and sea level pressure in each experiment onto the corresponding SST index (negative Niño-3.4, TA20, and IOBM; Fig. 6). The Pacific induces a high-pressure anomaly over Dumont d'Urville Sea (Fig. 6a, b), although with larger intensity and the center located west of that in the regression result using ERA5 data. The Atlantic Ocean induces a low-pressure anomaly over the Dumont d'Urville Sea and Wilkes Land

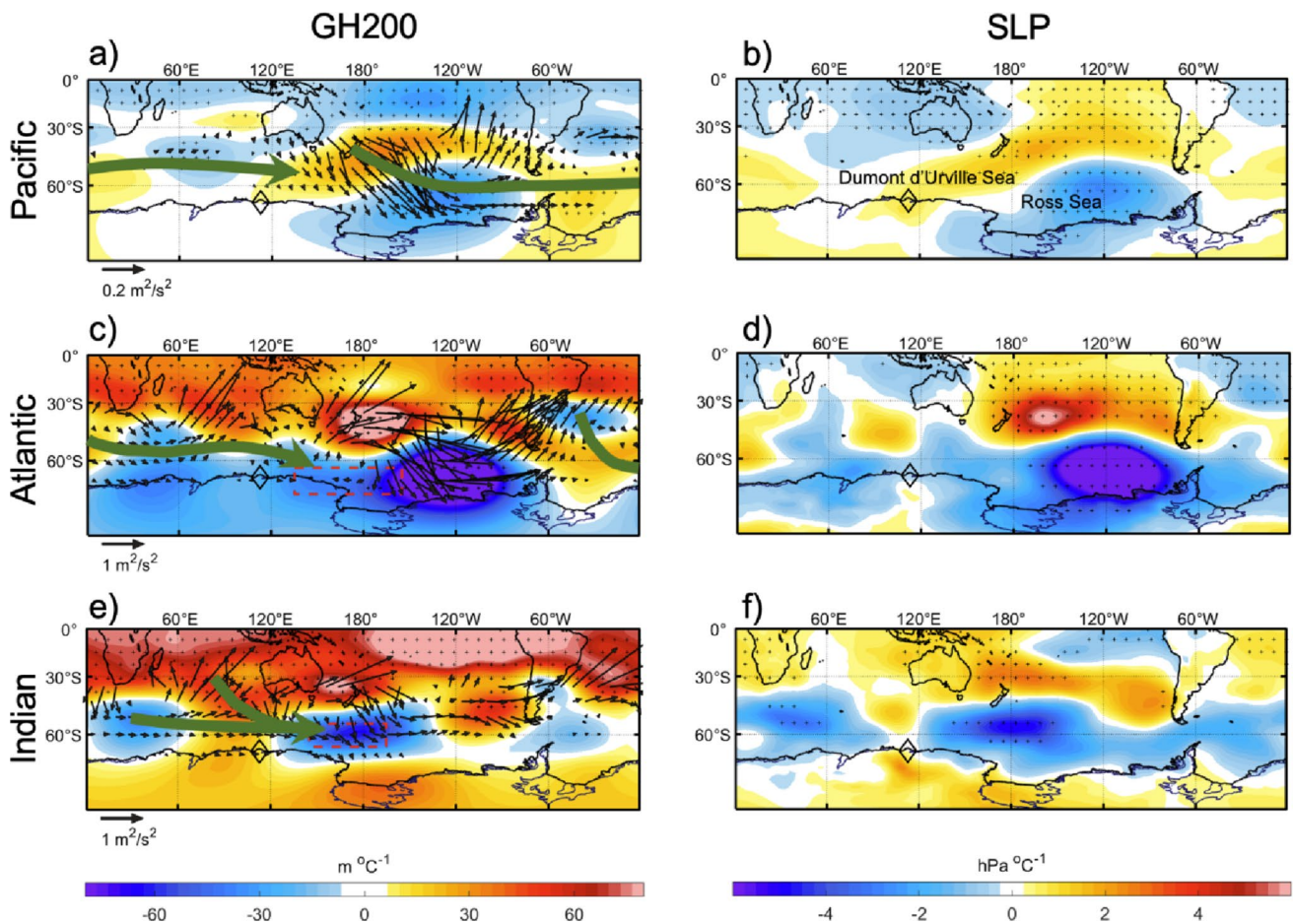


Fig. 5 Regression patterns of atmospheric circulation and the wave activity flux in ERA5 on the tropical SST indices. **a** shows the pattern of regression coefficients of the detrended geopotential height at the 200 hPa level (color shading) and the T–N wave activity flux (black vectors) on the tropical Pacific index, detrended negative Niño-3.4. Black crosses denote the areas where the regression coefficients of geopotential height are statistically significant at the 95% confidence level. **b** shows the regression pattern of sea level pressure on negative Niño-3.4. **c** and **d** show the regression pattern of geopotential height at the 200 hPa level (**c**) and sea level pressure (**d**) on the tropical Atlantic

index, detrended TA20. **e** and **f** show the regression pattern of geopotential height at the 200 hPa level (**e**) and sea level pressure (**f**) on the tropical Indian Ocean index, detrended IOBM. The red boxes in (**c**) and (**e**) outline the target regions used to define the local Z200 circulation indices for the quantitative assessment of the Rossby wave teleconnections. The schematic green arrows in panels (**a**), (**c**), and (**e**) illustrate the idealized Rossby wave propagation pathways originating from each tropical ocean basin. The open diamonds in all panels mark the location of the Law Dome ice-core site

(Fig. 6c, d), reproducing the atmospheric circulation pattern in the regression results using ERA5 data. The Indian Ocean induces a low-pressure anomaly over Dumont d’Urville Sea and a high-pressure anomaly over Wilkes Land (Fig. 6e, f). The low-pressure anomaly is weaker and more northerly than the regression result using ERA5 data and the high-pressure anomaly is stronger and more easterly. The associated wave flux shows that the atmospheric anomalies over Victoria Land and Dumont d’Urville Sea in the three experiments are all related to Rossby wave trains which are first stimulated in the subtropical Pacific, Atlantic and Indian Oceans and then propagate eastwards to Wilkes Land and Dumont d’Urville Sea (vectors in Fig. 6a, c and e). Overall, the experiments support the inference from ERA5 that tropical SST variability can excite extratropical wave trains

impacting East Antarctic circulation, while also highlighting basin-dependent differences in the intensity and longitudinal placement of the terminal anomalies.

We next examine the moisture-transport anomalies associated with the basin-dependent circulation adjustments (Fig. 7). Using ERA5, vertically integrated moisture transport is regressed onto each SST index (Fig. 7a, c and e). The high-pressure anomaly over Dumont d’Urville Sea associated with the eastern tropical Pacific cooling induces westerly winds over Wilkes Land and Victoria Land (vectors in Fig. 7a), with a weak influence on the moisture transport from the ocean to Wilkes Land (color shading in Fig. 7a). The low-pressure anomaly over Dumont d’Urville Sea induced by the warming in the Atlantic and Indian Oceans induces anomalous southerly winds (vectors in Fig. 7c, e),

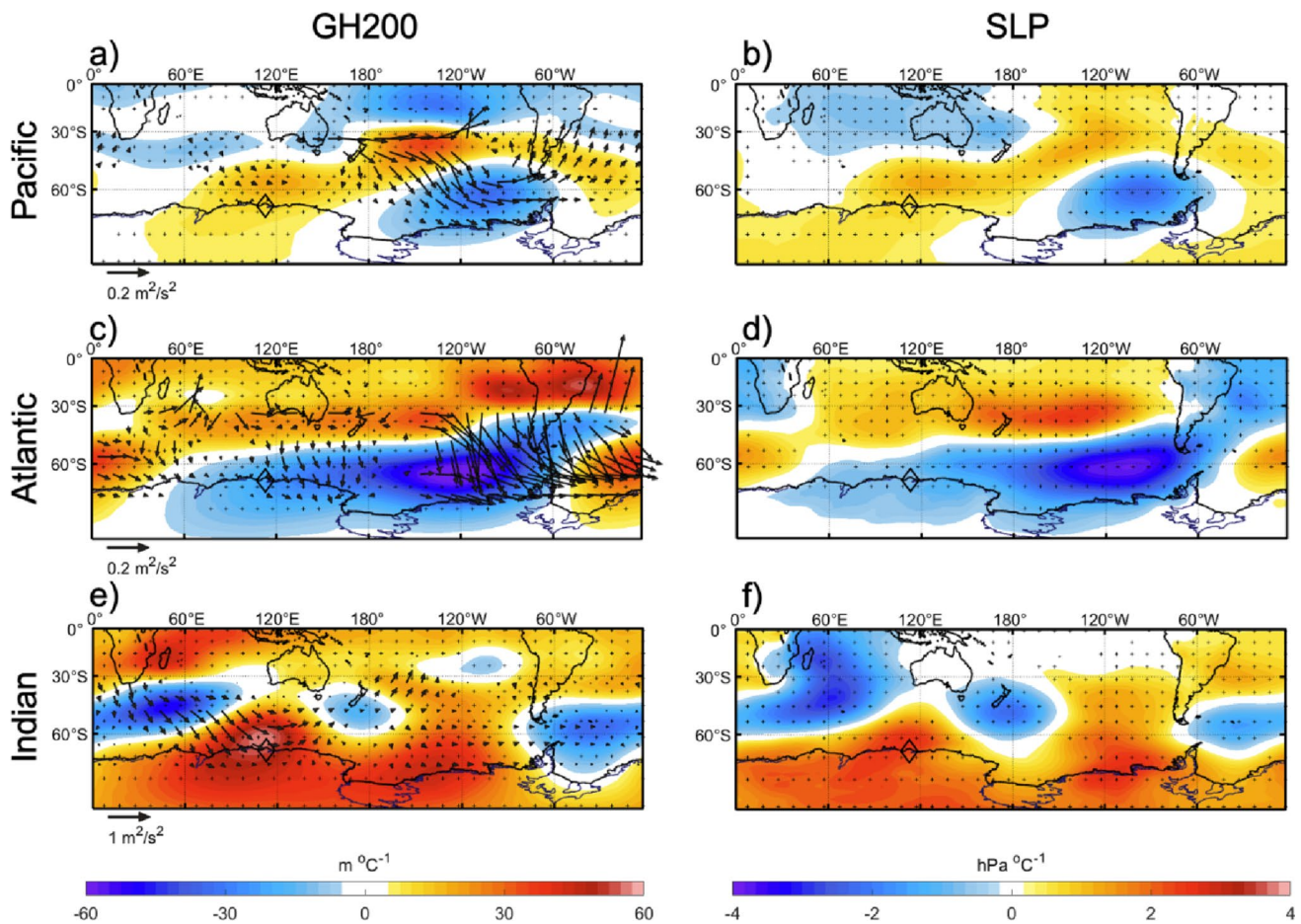


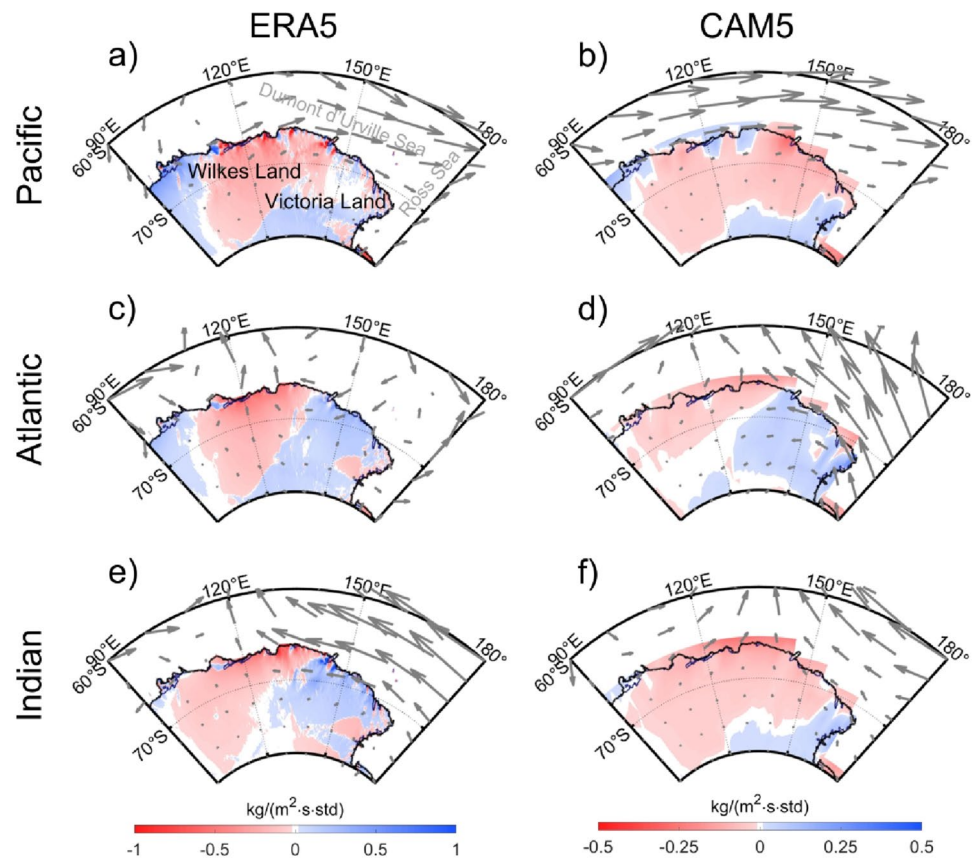
Fig. 6 Regression patterns of simulated atmospheric circulation and the wave activity flux from CAM5 on the tropical SST indices. **a** shows the pattern of regression coefficients of the detrended geopotential height at the 200 hPa level (color shading) and the T–N wave activity flux (black vectors) on the tropical Pacific index, detrended negative Niño-3.4. Black crosses denote the areas where the regression coefficients of geopotential height are statistically significant at

the 95% confidence level. **b** shows the regression pattern of sea level pressure on negative Niño-3.4. **c** and **d** show the regression pattern of geopotential height on 200 hPa level (**c**) and sea level pressure (**d**) on the tropical Atlantic index, detrended TA20. **e** and **f** show the regression pattern of geopotential height at the 200 hPa level (**e**) and sea level pressure (**f**) on the tropical Indian Ocean index, detrended IOBM

inducing anomalous moisture transports from the interior of Wilkes Land to the ocean (color shading in Fig. 7c, e). Consistent with the ERA5 results, the sensitivity experiments reproduce the basin-dependent moisture-transport patterns (Fig. 7b, d and f). Notably, the Pacific-forced experiment suggests locally enhanced onshore moisture transport in coastal regions (Fig. 7b), further indicating that the Pacific-driven circulation response is not conducive to the observed Wilkes Land SMB reduction. Taken together, these results indicate that the low-pressure anomalies over Wilkes Land and Dumont d’Urville Sea induced by warming in the tropical Atlantic and Indian Oceans inhibit the moisture transport from the ocean to the ice sheet, resulting in the decreasing SMB trends in Wilkes Land, while the cooling in the eastern tropical Pacific potentially induces high-pressure anomalies that are irrelevant to the reduced moisture transports from the ocean to Wilkes Land.

In summary, our multi-dataset and modeling framework establishes a complete physical chain: multi-decadal warming in the tropical Atlantic and Indian Oceans excites stationary Rossby wave trains that propagate to East Antarctica; these waves manifest as an anomalous barotropic low over the Dumont d’Urville Sea; and the resultant cyclonic circulation drives offshore winds that starve Wilkes Land of marine moisture, ultimately leading to the observed decline in SMB. Quantitatively, the Atlantic-forced Z200 index is negatively correlated with TA20 ($r = -0.30$, $p < 0.1$), and the Indian-forced Z200 index is negatively correlated with IOBM ($r = -0.34$, $p < 0.05$). In turn, both regional Z200 indices are positively correlated with Wilkes Land precipitation ($r = 0.56$ and 0.55 , respectively; both $p < 0.05$), indicating that a deeper regional low-pressure anomaly is associated with reduced precipitation over Wilkes Land.

Fig. 7 Regression patterns of moisture transports on the tropical SST indices. **a** shows the pattern of regression coefficients of the detrended vertically integrated moisture transports (vectors) and their projection on gradient direction of surface elevation (color shading) in ERA5 on the tropical Pacific index, normalized and detrended negative Niño-3.4. **b** shows the regression pattern of the detrended simulated vertically integrated moisture transports (vectors) and their projection on the gradient direction of surface elevation (color shading) from CAM5 on negative Niño-3.4. **c** and **d** show the regression pattern of moisture transports in ERA5 (**c**) and simulated moisture transports by CAM5 (**d**) on the tropical Atlantic index, normalized and detrended TA20. **e** and **f** show the regression pattern of moisture transports in ERA5 (**e**) and simulated moisture transports by CAM5 (**f**) on the tropical Indian Ocean index, normalized and detrended IOBM



4 Discussion

Our results reveal a clear link between multi-decadal tropical ocean variability and SMB changes over Wilkes Land, East Antarctica. This finding is consistent with previous studies showing that tropical SST variability can influence Antarctic climate through stationary Rossby-wave teleconnections (Li et al. 2021). However, our analysis further indicates that the dominant contribution to Wilkes Land SMB decline arises from warming in the tropical Atlantic and Indian Oceans, whereas the eastern tropical Pacific cooling plays a comparatively minor role. Previous studies have primarily emphasized the influence of the tropical Pacific on Antarctic climate variability, especially over West Antarctica and the Antarctic Peninsula through ENSO-related teleconnections (King et al. 2023; Lee et al. 2015; Li et al. 2021; Macha et al. 2025; Zinck et al. 2025). For example, Pacific variability has been shown to modulate atmospheric circulation patterns over the Amundsen–Bellingshausen Sea sector and influence precipitation and temperature variability in West Antarctica (Man et al. 2025; Bodart and Bingham 2019; Donat-Magnin et al. 2020). By contrast, our results indicate that the Pacific Ocean plays a relatively minor role in explaining the observed SMB decline over Wilkes Land. Instead, warming in the tropical Atlantic and Indian Oceans emerges as the dominant driver,

highlighting an important contrast between the Pacific-dominated variability in West Antarctica and the Atlantic–Indian influence identified here for East Antarctica.

The physical mechanism identified in this study is dynamically consistent with established frameworks of tropical–extratropical teleconnections (Ding et al. 2011; Li et al. 2015b; Simpkins et al. 2014). Warming in the tropical Atlantic and Indian Oceans excites stationary Rossby wave trains that propagate across the Southern Hemisphere midlatitudes and terminate near the Dumont d’Urville Sea. These wave trains induce a barotropic cyclonic anomaly off the coast of Wilkes Land, which modifies the regional wind field and suppresses onshore moisture transport toward the ice sheet. Moisture-flux–convergence control on Antarctic snowfall and the importance of topography-aligned (upslope) moisture transport for coastal precipitation provide a physically grounded link from the circulation anomaly to reduced precipitation (Bromwich 1988; Wille et al. 2019). This mechanism is supported by both regression diagnostics using reanalysis data and targeted atmospheric model experiments, lending confidence to the causal interpretation of the tropical forcing. The prescribed-SST CAM5 experiments are designed to isolate the atmospheric response to basin-scale SST anomalies and therefore provide strong support for a forcing contribution from the tropical Atlantic and Indian Oceans. However, these experiments do not represent coupled

atmosphere–ocean–sea-ice feedbacks, and the simulated response can be sensitive to the imposed boundary conditions and to the prescribed sea-ice state. We therefore interpret the experiments as evidence for a physically plausible and basin-specific forcing pathway, rather than as a complete attribution of all components of the observed Wilkes Land SMB trend.

The results also contribute to a growing body of literature suggesting that the Atlantic and Indian Oceans exert a stronger influence on Southern Hemisphere climate variability than previously recognized. Recent studies have shown that tropical Atlantic warming can generate large-scale atmospheric circulation responses extending into the Southern Ocean (Li et al. 2021; Simpkins et al. 2014), while Indian Ocean basin-wide warming has been linked to changes in the Southern Hemisphere jet and storm tracks (Ma et al. 2025, 2026). Our findings extend these insights by demonstrating that such basin-scale SST variability can also influence Antarctic surface mass balance through modulation of moisture transport pathways.

Despite the robust multi-dataset evidence presented here, several uncertainties remain. First, observational constraints on SMB variability and trends across Wilkes Land remain limited by the scarcity of long, continuous ice-core records, motivating additional records to better quantify spatial coherence and reduce sampling uncertainty (Favier et al. 2013; Gossart et al. 2019; Mottram et al. 2021; Wang et al. 2021). Second, while gridded products broadly agree on the sign of the SMB/precipitation decline, uncertainties in coastal East Antarctica arise from resolution limits and from imperfect representations of snow processes (e.g., drifting/blowing snow, sublimation) and orographic precipitation in reanalyses and models (Gossart et al. 2019; Hersbach et al. 2020; Mottram et al. 2021), as well as from structural differences among regional climate models (Agosta et al. 2019; Mottram et al. 2021; van Wessem et al. 2018). Third, the atmospheric modeling experiments employed here prescribe SST anomalies in individual ocean basins and do not fully account for coupled interactions among the atmosphere, ocean, and sea ice. In reality, tropical SST variability can interact with Southern Ocean processes (Meehl et al. 2016, 2019), potentially amplifying or modifying the atmospheric teleconnection pathways identified here (Krinner et al. 2014; Wang et al. 2020a).

Importantly, the identified Atlantic–Indian pathway provides a dynamical explanation for the strong regional heterogeneity of SMB trends within East Antarctica, suggesting that multi-decadal shifts in tropical SST patterns can modulate coastal precipitation through circulation-driven moisture supply and potentially offset thermodynamic expectations of increased snowfall in a warmer atmosphere (Frieler et al. 2015; Palerme et al. 2017; Tewari et al. 2022; Man et al. 2026). Future work combining additional ice-core constraints, higher-resolution regional modeling, and fully

coupled experiments will be needed to better quantify this sensitivity and to clarify the role of atmosphere–sea ice–ocean feedbacks in shaping precipitation and SMB patterns in East Antarctica.

5 Conclusion

This study provides multi-dataset evidence for a multi-decadal decline in SMB over Wilkes Land, East Antarctica, during 1979–2012. The decline is consistently indicated by the Law Dome ice-core record, regional climate model reconstructions, and ERA5-derived net precipitation, and is primarily associated with reduced precipitation rather than changes in other SMB components. These results highlight that East Antarctic SMB trends can be strongly regionally heterogeneous, and that local SMB changes may depart from the broad thermodynamic expectation of increased Antarctic accumulation under warming.

By combining observational diagnostics with basin-specific CAM5 experiments, we show that the Wilkes Land drying is most strongly linked to warming in the tropical Atlantic and Indian Oceans, whereas the contemporaneous cooling in the eastern tropical Pacific plays a comparatively limited role. The Atlantic and Indian Ocean warming is associated with Rossby-wave responses that project onto a barotropic cyclonic anomaly near the Dumont d'Urville Sea and Wilkes Land. This circulation anomaly suppresses onshore moisture transport toward the ice sheet, thereby reducing precipitation and SMB over Wilkes Land.

These findings identify an Atlantic–Indian tropical teleconnection pathway as an important contributor to East Antarctic hydroclimate variability on multi-decadal timescales. More broadly, they suggest that regional SMB changes over the EAIS cannot be understood from local thermodynamic effects alone, but also depend on how tropical SST anomalies project onto Southern Hemisphere circulation and coastal moisture-transport pathways. This has implications for future projections of Antarctic mass balance: climate models need to realistically represent basin-specific tropical SST variability and its teleconnection to East Antarctica in order to capture regional SMB changes over sectors such as Wilkes Land.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00382-026-08251-8>.

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Author contributions Kai Man, Xichen Li: Conceptualization, Formal analysis, methodology, writing—original draft, writing—reviewing

and editing. Yetang Wang, Lei Geng: Data curation, writing—reviewing and editing. Guitao Shi, Yonghao Wang, Yujie Miao and Weihang Ma: writing—reviewing and editing.

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Data availability Met Office Hadley Centre sea ice and sea surface temperature dataset (HadISST) is available at <https://www.metoffice.gov.uk/hadobs/hadisst>. The European Centre for Medium-Range Weather Forecasts Reanalysis version 5 (ERA5) is available at <http://cds.climate.copernicus.eu/datasets>. Surface mass balance data from RACMO2 are available at <https://www.projects.science.uu.nl/iceclimate/publications/data/2018/index.php>. Surface mass balance data from MARv3 are available at <https://doi.org/10.5281/zenodo.2547637>. BedMachine Antarctica, version 3, is available from the NASA National Snow and Ice Data Center Distributed Active Archive Center (<https://nsidc.org/data/nsidc-0756/versions/3>). Further data analysis and figures are made with MATLAB version R2018a, accessible at <https://in.mathworks.com/products/matlab.html>. The Community Atmosphere Model version 5 (CAM5), used in this study, is distributed as part of the Community Earth System Model (CESM) and is openly available via the CESM/CAM source-code repositories and download instructions (<https://github.com/ESCOMP/CESM>). The ice-core data, the model outputs generated and analysed in this study, and the statistical analysis code are publicly available from Zenodo at <https://doi.org/10.5281/zenodo.18877297>.

Declarations

Conflict of interest The authors declare that they have no conflict of interest or competing interest.

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